

Passive photodiode visibility — run report

RUN

190437_pd-visibility_pd-visibility

Status: completed | Started: 2026-09-23T18:04:37.051681+00:00

PURPOSE

Measure fringe contrast on the selected PD input under external drive. No RP output initialization or PAX connection.

REQUIRED OPTICAL SETUP

DC-coupled PD on the selected RP input; external drive must span complete fringes. Set its frequency and, when known, the blocked-light voltage offset. RP must already have the Pyrpl FPGA loaded.

RECORDED BENCH CONTEXT (RECIPE)

PD: in1; external drive: 0.5 Hz; RP outputs unchanged

OPERATOR SETUP NOTES

V_{pi} expectation only; verify channel and measured driver gain before calibration.

DATA

3 recorded samples; acquisition: 3

Source: data.csv; full acquisition settings: recipe.json; status: run.json

Photodiode visibility

PD INPUT

in1

CAPTURES

3

VALID CAPTURES

3

VISIBILITY MEAN

0.012563069555076854

VISIBILITY STD

0.0023375497669405176

DARK VOLTAGE V

None

EXTERNAL FREQUENCY HZ

0.5

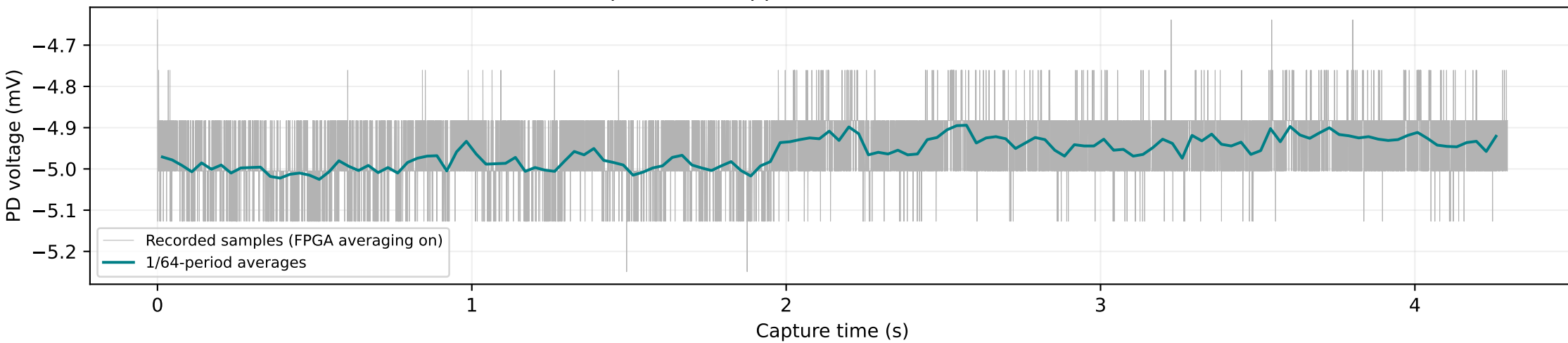
DEFINITION

$(I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$, $I = \text{polarity} * (V - V_{\text{dark}})$; extrema of 1/64-period averages

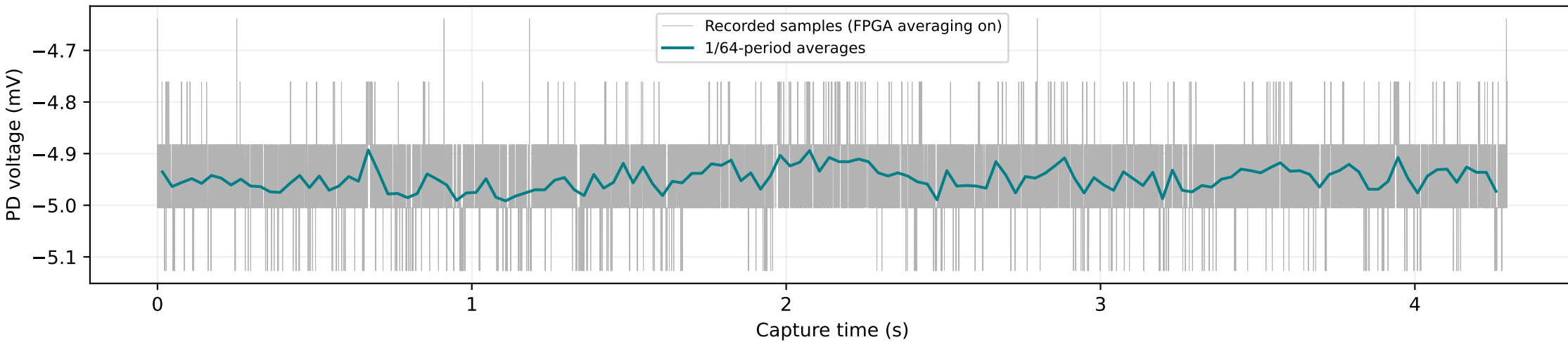
INTERPRETATION

Observed contrast assumes DC coupling, linear unsaturated PD, and full fringe excursion. A missing dark measurement gives apparent contrast only. Scatter is not total uncertainty.

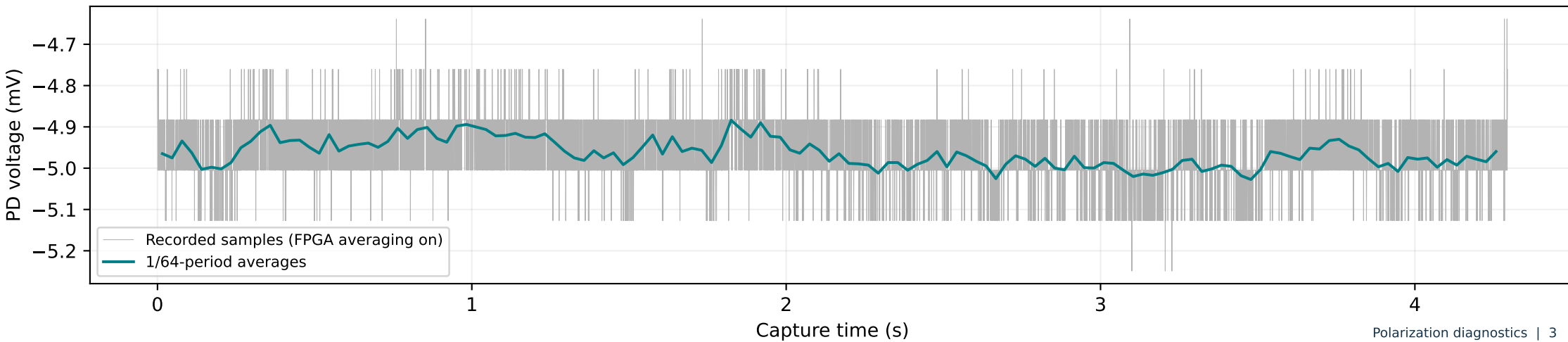
in1 / capture-000: apparent; zero dark offset assumed



in1 / capture-001: apparent; zero dark offset assumed



in1 / capture-002: apparent; zero dark offset assumed



Active parameters

Passive acquisition; voltage at the selected PD input. No actuator commands.

Parameter	Recorded value
label	"pd-visibility"
duration_s	12.0
report	"pdf"
pd_input	"in1"
visibility_frequency_hz	0.5
visibility_dark_voltage_v	null